08-15 第2篇

阅读理解（说明文） Why Does the Evening Sky Turn Red?

① Almost every child has asked the same question at least once: "Why is the sky blue in the afternoon but red when the sun is setting?" It is a simple question, but the full answer involves some fascinating science about **light**, **air**, and **distance**. Once you understand it, you will look up at every sunset with a new sense of wonder.

② The first thing to know is that white **sunlight** is not really "white" at all. It is a mixture of many colours. If you have ever seen a rainbow, you already know this. Rain drops act like tiny glass prisms; they break the sunlight into seven main colours: red, orange, yellow, green, blue, indigo, and violet. Each colour travels as a wave, and each colour has a slightly different **wavelength**. Red has the longest wavelength; blue and violet have much shorter ones.

③ The second thing to know is that our air — the **atmosphere** — is not as empty as it looks. It is full of tiny gas particles, especially **nitrogen** and **oxygen**. When sunlight travels down through this thick layer of gas, the small particles get in the way. They **scatter** the light in many directions, like a thousand small torches flashing at the same time.

④ Here is the important point: the tiny particles do not scatter all colours equally. **Shorter wavelengths are scattered much more strongly than longer ones.** Blue light, because its wavelength is short, is thrown in every direction by the air. Red light, with its long wavelength, goes through the air more easily in a straight line, and is scattered far less.

⑤ In the middle of the day, the sun is high in the sky. Sunlight has a fairly short path to travel through the atmosphere. Blue light is scattered so widely that it seems to come from every part of the sky above us. That is why the afternoon sky looks **blue** — we are seeing the scattered blue light everywhere.

⑥ Near sunset, the situation changes completely. The sun is low on the horizon. Its light now has to pass through a much **longer path of air** — sometimes many times longer than at noon — before reaching our eyes. On this long journey, almost all the blue light is scattered away before it can get to us. Only the longer red and orange waves manage to continue in a straight line, reach our eyes, and paint the clouds and the whole western sky in warm colours.

⑦ This also explains why a sunset looks especially bright and deep red on days with much **dust** or **smoke** in the air. Larger particles scatter even more of the shorter colours,

leaving an even purer red. So the most beautiful sunsets often appear after a dry, dusty day, or sometimes, sadly, on days with a lot of pollution.

⑧ Next time you stand at your window in the evening and watch the red sky slowly fade behind the buildings, you will know that what you are seeing is really a simple physics lesson: a long journey of light, and a last gift from the sun before it sets.

注释： wavelength 波长；atmosphere 大气层；nitrogen 氮气；oxygen 氧气；scatter 散射

根据文章内容，选择最佳答案。

1. What is the main thing the writer wants to explain in this passage?

A. Why rainbows appear after the rain.
B. Why the sky looks blue at noon but red at sunset.
C. How to paint a beautiful sunset with a brush.
D. Why air pollution is increasing in our cities.

2. According to the passage, which colour of light has the longest wavelength?

A. Blue.
B. Green.
C. Violet.
D. Red.

3. Why does the afternoon sky look blue?

A. Because the air is painted blue by the sea.
B. Because only red light is able to pass through the air in the afternoon.
C. Because the tiny gas particles in the air scatter blue light strongly in every direction, so we see blue from every part of the sky.
D. Because the sun itself is blue in the middle of the day.

4. Why does the sunset sky look red?

A. Because the sun burns harder in the evening.
B. Because at sunset sunlight passes through a much longer path of air, and almost all the blue light is scattered away, leaving mainly red and orange light to reach our eyes.
C. Because red light is heavier than blue light.
D. Because the sky is always red at night.

5. What can we learn from the last paragraph?

A. Sunsets are boring natural events with no meaning.
B. A red sunset is basically a simple physics lesson about the journey of light.

C. We should never watch sunsets, because they hurt our eyes.
D. Sunsets only appear in cities with no tall buildings.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】第①段问题及第⑤、⑥段解答
【解析】文章围绕"为何正午天蓝而日落天红"展开。A 项只是第②段举例；C、D 与文意无关。故选 B。

2.
【答案】D
【题型】细节理解题
【定位】第②段末 "Red has the longest wavelength"
【解析】原文明确指出红光波长最长。A、B、C 均不符。故选 D。

3.
【答案】C
【题型】细节理解题
【定位】第⑤段 "Blue light is scattered so widely that it seems to come from every part of the sky above us"
【解析】正午天蓝因蓝光被大气粒子向四面八方散射。A、D 属无中生有；B 项与"正午短路径、蓝光散射强"逻辑相反。故选 C。

4.
【答案】B
【题型】推理判断题
【定位】第⑥段 "longer path of air ... almost all the blue light is scattered away ... Only the longer red and orange waves ... reach our eyes"
【解析】日落时光经过更长空气路径，蓝光几乎全被散射，剩余红橙光到达人眼。A、C、D 均无文中依据。故选 B。

5.
【答案】B
【题型】写作意图题
【定位】第⑧段 "a simple physics lesson: a long journey of light"
【解析】作者结尾把日落视为光的旅程与简单物理课。A、C、D 均与作者态度相反。故选 B。